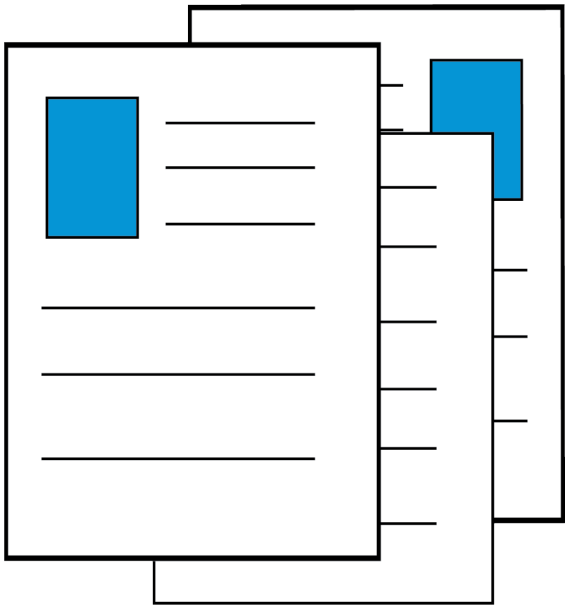
An Agile Transformation Case Study On Wiredrive



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Executive Summary

In 2008, Wiredrive faced two major problems that had been years in the making. The first was that it had two competing business focuses: custom Web-design; and online file-sharing services. Both provided revenues, but while the long-term potential of the latter was larger, the attention devoted to the former slowed progress on the file-sharing service.

The second was difficulty in planning and implementing new capabilities for the file-sharing service. The company's "Waterfall" process worked, but its pace and flexibility proved disappointing.

Motivated by the second problem, the company decided to migrate from their existing Waterfall process, to Scrum. The year-long transition succeeded because all company stakeholders bought in to the need and sustained the commitment over the time required to work through the challenges.

Wiredrive's new Scrum process increased visibility into all aspects of software development. Surprisingly, this visibility brought new clarity to business issues, as well as to development priorities. As a result, the company decided to re-write the file-sharing application to improve flexibility, and to terminate the custom Web design service in favor of the file-sharing business.

Wiredrive adopted Scrum as a tactic to improve the company's ability to develop software, but found that it also enabled strategic business decisions that provided major benefits. As a result, Wiredrive considers the transition to Scrum to be a success.

Introduction

Wiredrive provides an online file-sharing service used by creative professionals in the advertising, television, motion-picture, and interactive industries. By 2010, tens of thousands of Wiredrive users had uploaded over 3,000,000 files and delivered over 2,000,000 presentations.

A key element in the success of Wiredrive is the company's agile software development process. The company's Scrum process has proved effective, but the transition to it was not easy. In this paper, we will look at the issues the company faced, how the company overcame the issues, and how the transition has benefited the company.

The Problems

In 2008, Wiredrive's problems with business focus and software development had combined to slow the company's progress to a crawl.

Business Focus

In the early years, Wiredrive provided a custom Web-development service, in addition to the file-sharing service.

The split focus impaired the company's ability to grow revenues. While the Web-development work brought in money on a time-and-materials basis, the file-sharing business provided a continuing revenue stream that was not restricted by billable hours. The file-sharing business was more profitable, but resource contention from the other service crippled its growth.

Software Development

The second problem area was in software development for the file-sharing service. The company was hampered by two major problems.

Technical Debt from Legacy Code

Several years of development work grew the file-sharing codebase dramatically from its initial implementation. Under constant pressure to add new capabilities, the development team did so at the expense of design. The result was a codebase whose design evolved more by accretion than intent.

The haphazard accumulation of "bolted on" code caused fragility. Poor design made adding new features difficult, and led to numerous defects, often in areas that seemed unrelated. Even worse, the company was unable to ensure quality before releasing the service to customers, and experienced an unacceptable level of complaints.

Unsuccessful Development Process

Wiredrive used the common "Waterfall" process to plan and execute software development. A typical Waterfall process contains the following major stages:

1. Requirements: Product Managers write requirements specifications
2. Design: Software architects or senior engineers write design documents that describe how the requirements are to be implemented
3. Execution
   a. Software Engineers write the software, to the design specifications
   b. QA Engineers write test cases to validate implementation of the requirements
4. Testing and Fixing
   a. QA Engineers test the completed application, and report defects
   b. Software engineers fix defects
   c. The test-fix cycle is repeated until the defect level is acceptable for release
5. Release: IT Operations deploys the application for customer use

   Most Waterfall processes share these characteristics:

   a. The project plan includes a schedule. The schedule is based on effort estimates derived from the requirements, and the availability of resources.
   b. The scope of requirements is large enough for the planned schedule to take several (3) to many (9+) months to complete.

While the process was well-defined and seemed reasonable on paper, it proved unsatisfactory in practice. The company could not deliver the desired features on anything resembling the planned schedule. Projects that were expected to take three months might take as long as eighteen.

Many factors contributed to the unsatisfactory performance.

1. Distractions from the custom Web-development service interfered with company and development focus

2. Requirements were often too big and ambitious for implementation to be practical
3. Requirements were often not well-enough defined to support estimation and implementation, and clarification caused churn that impacted the schedule
4. Even when requirements were understood reasonably well, estimates were imprecise, and work did not proceed according to the schedule
5. Scope creep and change requests disrupted the planned work, and delayed its completion
6. Software defects generated high-priority customer requests for fixes, which diverted resources from development work

As a result, the company was unable to plan and implement features in a meaningful way. Wiredrive's customers perceived the company as unresponsive to their needs, unreliable in its promises, and unable to provide acceptable quality.

Solutions

The problems Wiredrive faced were well understood within the company. What was missing was the ability to resolve them. Starting in 2008, the company focused its efforts on resolving these problems by making two major changes.

Solving the Focus Problem

The solution was straightforward, in principle: Drop one line of business and focus effort on the more profitable one. Wiredrive's challenges were in recognizing the degree to which the split focus was slowing progress, and having enough revenues to be able to drop the less-valuable business.

In 2008 and 2009, the company's effort to improve their development process improved visibility not only into development, but with respect to business priorities. It became clear that progress in the file-sharing area could only occur if the custom Web-development business was terminated.

At about this same time, the company's financial position became secure enough that it was able to make the decision to close the Web-development business. This closure enabled the company to focus on the more profitable file-sharing business, and complete its migration to an agile development process.

Solving the Software Development Problem

Wiredrive's President, Bill Sewell, began hearing about the Scrum process for agile development around 2004. The principles of agile development sounded good. As the Agile Manifesto put it,

> ***We value Individuals and interactions over processes and tools***
>
> ***Working software over comprehensive documentation***
>
> ***Customer collaboration over contract negotiation***
>
> ***Responding to change over following a plan***
>
> ***That is, while there is value in the items on the right, we value the items on the left more***
>
> \- *www.agilemanifesto.org*

Bill liked the emphasis on people and results over process, and the idea that the company could become more responsive to customers. One particular agile process, Scrum, seemed particularly appropriate for the company's needs. The challenge was in making the change happen. The Scrum process defined a particular set of practices that promised to help the company achieve this new agility, but at the cost of radically changing almost everything about how Wiredrive planned and executed its software development.

The Beginning

The first step towards a Scrum process occurred in October 2008, when Stefan Leikin, who managed development of the file-sharing software, attended a Certified Scrum master class. Stefan not only attended the class, but arrived early, stayed late, and took lunch-time opportunities to talk with instructor Scott Downey. Scott taught Scrum and served as Head Agile Coach for MySpace.com. He provided a wealth of practical experience in his conversations with Stefan.

Stefan was sufficiently impressed with Scott that Wiredrive invited Scott to do some on-site training for members of the engineering team. Before leaving, Scott asked the students if they were excited about Scrum, or dreading it. Stefan recalls that everyone was "pumped" at the prospect of using Scrum to manage their development work.

The Plunge

One advantage Wiredrive had in making its transition to Scrum was its size: The company had fewer than forty people, and only six in the development group. Bill, Stefan, Taylor Tyng (the Product Manager), the CTO, and the development team members were unanimous in believing that Scrum was the right choice.

By 2009, all of the stakeholders had committed to the Scrum process, and the company conducted its first Scrum "Sprint."

The Transition

The major challenges faced by the development team were in three areas: Mastering Scrum, Keeping Commitments, and Managing Technical Debt.

Mastering Scrum

While the practices of Scrum are not complex, most are unfamiliar, and learning the full set is a challenge. Scrum projects repeat their "cadence" of activities every Sprint (development cycle, two weeks for Wiredrive). Each activity must occur at its specified time, because the process is tightly choreographed. Missing any step can impact the team's productivity dramatically.

Successful Scrum projects must go beyond learning the practices to master the choreography, until it becomes second nature. For Wiredrive, this degree of mastery occurred after about eight Sprints.

Keeping Commitments

Each time the development team planned a Sprint, the team committed to a specific scope. This commitment was possible because the team knew from past experience how much work it could do in a Sprint, and estimated the amount of work needed to implement the selected requirements.

Wiredrive's metric for a Sprint's success was whether the team implemented the planned scope. By this definition, the team had its first success in Sprint 13. This success was followed by eight failures. Results improved gradually, until by Sprint 30 they'd had five successful Sprints in a row.

Managing Technical Debt

The fragility of Wiredrive's software had reached the point where further development was impractical. The development team advocated "going dark" to re-design and re-implement the software, before trying to add more features. Bill describes himself as the last person who wanted to hear about the need to "go dark," and not "shoehorn 2—3 more features to get us out in

front of the marketplace," but the team found that attempts to add more features kept breaking the application. After much discussion, the company decided to address the technical debt.

Wiredrive did "go dark." The development team implemented a clean-sheet design to provide a platform for future development, and created extensive automated tests to enable quick assessment of product quality. This process took months, during which the company provided no new features to customers. The decision paid off, in terms of software stability and quality, and the company has been able to resume feature development with much less breakage.

The Results

Now past the one-year mark, Wiredrive's experiment with Scrum is universally viewed as a success within the company. The company and its customers have observed many benefits:

- The development team can plan work, and execute to the plan, routinely
- The minimum time-to-market for feature requests is down from approximately one year to four weeks.
- The number of serious defects reported by customers, per release, is down
- Scope creep and change requests have completely disappeared
- Frustration has decreased, and morale has improved, throughout the company
- Customers are happier with the company's responsiveness and product quality

Major Findings from the Scrum Transition

The benefits of a Scrum process are usually described in terms of customer satisfaction, which is important. However, Wiredrive learned some interesting lessons from this transition, beyond issues of customer satisfaction. These findings are described below.

The Importance of Visibility

It is often said of military planning, that, "Amateurs study strategy. Professionals study logistics." One implication of this statement is that the most important things in a particular domain are not always the most obvious.

The same dictum holds true for Wiredrive's discoveries about Scrum. The company found that the Scrum process.

- Compressed schedules into short Sprints, in which it became obvious very quickly whether the work was on track
- Tracked work at a fine-grained task level, daily, so that the current status of development was always clear
- Forced explicit trade-offs between conflicting desires for feature development

In short, the most important benefit of the Scrum process was one that made all of the others possible: Visibility.

The development team members and other stakeholders quickly discovered that issues which had remained hidden, or could be deferred, in the past, were now forced into view. The discoveries were often painful, because they required immediate action for progress to be possible. Yet it was exactly this process of discovering and addressing painful issues that made longer-term success possible.

Bill has made this observation about the change.
"In a world that is completely uncertain all the time, we can do the best job possible with the information we have control over, and we can roll with the rest."
Stefan agrees, adding that transparency reduces anxiety, and the 'useless stupid stuff' that gets in the way of the company.

"Scrum doesn't make everything perfect, but lays it out on the table."
The visibility provided by Scrum played a pivotal role in other lessons learned.

The Need to Focus the Business

It was during the build-up to Scrum that the company realized how its split focus hampered progress in the more profitable file-sharing business. Terminating the custom Web-development offering was necessary if the company was to achieve its goals. While this decision was not a consequence of an existing Scrum process, it flowed naturally from the increased focus on fundamentals (visibility) that grew during the transition to Scrum.

The decision was painful, but necessary, and the benefits from the new focus followed quickly. "It was like watching the clouds part," as Stefan put it. Distractions vanished, and the company could finally devote its efforts to the area that made sense.

The Urgency of Retiring Technical Debt

Visibility played a major role in the decision to 'retire the technical debt' by re-writing the file-sharing codebase. The application's fragility, and the lack of automated testing, made progress in short development cycles impossible. Where the previous process allowed the company to push the debt down the road, the Scrum process forced the company to make a decision about the debt.

Wiredrive decided to place a hold on new-feature development, and invest the time needed to re-write the application and develop automated-test capabilities. This decision paid off, by enabling new-feature development without the fragility that had plagued the application in the past.

The Value of Persistence

Success did not come overnight. Wiredrive's development team required 7—8 sprints to become comfortable with Scrum, and approximately 25 before they could routinely complete work as planned. This transition took approximately one year.

During this year, the company encountered many obstacles, such as difficulty in mastering the basics of the Scrum process, coping with technical debt, and adjusting their methods to work with a remote team member. Ultimately, it was the company's persistence that made the transition to Scrum possible. This persistence was driven by the belief that Scrum would improve their ability to perform, and backed by commitment from all stakeholders.

In contrast, half-hearted support and lack of shared vision among stakeholders would likely have doomed the process, leading to the worst of both worlds: A failure to improve, and wasted effort from the attempt. Wiredrive avoided this fate by achieving stakeholder buy-in across the company before starting the migration to Scrum.

The Lessons of Failure

The Scrum process emphasizes learning from experience, by making Retrospective meetings part of the process. The lessons learned relate mostly to tactical and developmental issues, and help the development team to improve over time.

Wiredrive also learned a strategic lesson about Scrum, which is that it is not always the right process. The company attempted to implement Scrum in their sales process, and found that the attempt caused

problems while providing little benefit. The plan for each Sprint included defining which clients to call, a concept that seemed reasonable, but which led to achieving only 20% of their goals, and failing to follow up on half of the sales leads.

After five iterations, the company evolved a "Swarm" strategy, which relies on tracking sales stages instead of Scrum tasks. This approach has proved to be more effective for the sales process.

Tools and Techniques

With a small group of six people in one office, Wiredrive was able to start with a co-located team, and use whiteboards and corkboards to plan and track progress.

Planning

The main tools for Planning are index cards.

Taylor, the Product Manager, is responsible for writing requirements in the Scrum "Story" format. He provides these Stories on index cards.

The development team kicks off each Sprint by estimating the size of the top-ranked Stories, selecting a set to be implemented in the Sprint, and creating a list of tasks (the task breakdown) for each Story. The task breakdown contains all tasks that are large or significant enough to be worth tracking, and whose execution implements the requirements. The team members write these tasks on index cards.

Tracking

Tracking and Planning are closely related, as both involve the same set of index cards. The Story and Task cards are tacked onto a corkboard, which is divided into columns labeled "Not Started," "In Process," and "Done." During Sprint execution, the team member who starts or completes a task moves the task card to the appropriate column. Thus at any time, the "Scrum board" displays the current status of all tasks planned for the Sprint.

Collaboration

Scrum emphasizes collaboration and self-organization among team members, both of which occur constantly at Wiredrive. Team members call their own meetings, hold impromptu whiteboard design sessions, and do pair programming all on their own initiative.

Tools used for collaboration at first included only whiteboards, corkboards, and the usual office applications (email, spreadsheets, word-processor documents). After the first several Sprints, however, one of the team members moved to Morocco, and the company was faced with the need to make Scrum work in a distributed environment.

Three key changes made distributed development workable.

1. The requirement for the remote team member to keep "California hours," working at the same time as the rest of the team.
2. The use of an inexpensive "electronic Scrum board," Scrumy.com, to track requirements and work status. Use of the latter resulted in some duplicate effort, as the same tasks are tracked on both the corkboard and Scrumy.com, but the company has found this degree of redundancy acceptable to date.
3. The use of electronic communications tools, such as iChat and conference calls, to provide as much real-time communication between distributed personnel as possible.

Conclusion

Wiredrive's transition to a Scrum development process was successful. The company is able to plan and execute software development more reliably than it could before, and while becoming more responsive to customer requests. Nine-month development cycles are thing of the past, along with the technical debt and split focus that hampered the company's ability to move forward.

More than anything else, the key contribution of the Scrum process was to make visible all of the problems, constraints, and trade-offs that could previously be ignored or deferred. The new visibility brought discomfort, as the company had to make a number of painful decisions in short order, but the benefits have been profound.